2D Matrices: Stretches and Enlargements

Questions 1–13

Question 1. Describe the following linear transformations:

a) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$

d) $\begin{pmatrix} 10 & 0 \\ 0 & 20 \end{pmatrix}$

e) $\begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$

Question 2. The matrix $\mathbf{A}$ represents an enlargement, centred at the origin, scale factor 2.

a) Write down the matrix $\mathbf{A}$.

b) Find $\mathbf{A}^2$.

c) Find $\mathbf{A}^3$.

d) Find $\mathbf{A}^{15}$.

Question 3. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$. Write down the area of the shape's image.

Question 4. A shape with area 4 is mapped to its image by the matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$. Write down the area of the shape's image.

Question 5. The triangle T has vertices at $(-2, 0)$, $(0, 9)$ and $(4, 0)$. The matrix $\mathbf{A} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ maps T to the triangle T'. Find the coordinates of the vertices and area of the triangle T'.

Question 6. Write down the matrix that represents each of the following linear transformations:

a) Stretch, parallel to the x-axis, scale factor 9.
b) Enlargement, centred at O, scale factor 15.
c) Stretch, parallel to the y-axis, scale factor $\frac{2}{3}$.
d) Stretch, parallel to the x-axis, scale factor 6, and a stretch, parallel to the y-axis, scale factor 2.

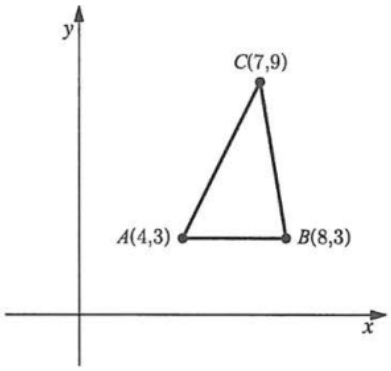
Question 7. Write down the matrix that represents a stretch that maps:

a) $(1, 2)$ to $(1, 4)$
b) $(4, -1)$ to $(12, -1)$
c) $(16, 9)$ to $(16, 3)$
d) $(25, -2)$ to $(5, -2)$
e) $(8, -7)$ to $(8, -3)$
f) $(-30, 5)$ to $(-18, 5)$

Question 8. Write down the matrix that represents an enlargement about the origin that maps:

a) $(2, 5)$ to $(8, 20)$
b) $(-240, 90)$ to $(-48, 18)$
c) $(20, 600)$ to $(64, 1920)$
d) $(-360, -440)$ to $(-342, -418)$

Question 9. The triangle T has vertices at $A(4,3)$, $B(8,3)$ and $C(7,9)$, as shown in the diagram below.



a) Find the area of T.

b) T is mapped to triangle V by the transformation associated with the matrix $\begin{pmatrix} 5 & 0 \\ 0 & 12 \end{pmatrix}$. Find the vertices of V and hence find its area.

Question 10. The kite K has vertices at $(0,-5)$, $(-1,0)$, $(0,2)$ and $(1,0)$. K is mapped to K' by an enlargement about the origin, such that the area of K is doubled. Find the exact coordinates of the vertices of K'.

Question 11.

a) Write down the 2×2 matrix that represents a stretch, parallel to the x-axis, scale factor 7, and a stretch, parallel to the y-axis, scale factor 3.

b) The point $(2m + n, n - m)$ is transformed by the matrix in a) to $(18n + 1, m - 1)$. Find the values of m and n.

Question 12. The point $A(-2, 8)$ is enlarged, centred at $B(2, -6)$, by scale factor 3. Find the coordinates of the image of the point A.

Question 13. The point $A'(10, -1)$ is the image of A after it is enlarged, centred at $B(-8, -3)$, by scale factor $\frac{1}{10}$. Find the coordinates of the point A.